19-202-0702 ADVANCED ARCHITECTURE AND PARALLEL PROCESSING

Course Outcomes:

On completion of this course the student will be able to:

1. *Summarize multiprocessors and multicomputer.*
2. *Utilize message passing mechanisms.*
3. *Outline memory hierarchy and caching mechanisms.*
4. *Elaborate pipelining and parallel programming.*

Module I

Parallel computer models-state of computing, multiprocessors and multicomputer, multivector and SIMD computers, PRAM and VLSI models, architectural development tracks. Program and Network Properties-Conditions of Parallelism, Program Partitioning and Scheduling, Program Flow Mechanism, System Interconnect Architectures. Principles of Scalable Performance-Performance Metrics and Measures, Parallel Processing Applications, Speedup Performance Laws, Scalability Analysis and Approaches.

Module II

Processors and Memory Hierarchy-Advanced Processor Technology, Superscalar and Vector Processors, Memory hierarchy technology, virtual memory technology. Bus, Cache and Shared Memory-Bus Systems, Cache Memory Organizations, Shared-Memory Organizations, Sequential and Weak Consistency Models.

Module III

Pipelining and Superscalar Techniques-Linear Pipeline Processors, Nonlinear Pipeline Processors, Instruction Pipeline design, Arithmetic Pipeline Design. Multiprocessor and Multicomputer-Multiprocessor System Interconnects, Cache Coherence and Synchronization Mechanisms, Message Passing Mechanisms. Vector Processing Principles.

Module IV

Parallel Programming- Parallel Programming Models, Parallel Languages and Compilers. Instruction Level parallelism- Design issues, model of typical processor, compiler-directed instruction level parallelism, operand forwarding, Tomsula’s algorithm, branch prediction, thread level parallelism.

References:

1. Kai Hwang, Naresh Jotwani, Advanced Computer Architecture Parallelism, Scalability, Programmability, 3rd edition, McGraw-Hill Education,2011
2. Dezso Sima, Terence Fountain, Peter Karasuk, Advanced Computer Architecture-A design space approach, Pearson Education,2012
3. Sajjan G Shiva, Advanced Computer Architecture, CRC Taylor & Francis,2006.
4. David E Culler, Jaswinder Pal Singh, Anoop Gupta, Parallel Computer Architecture, Elsevier.
5. Hwang and Briggs,Computer Architecture and Parallel Processing,McGraw Hill.